

===== Banking Application =====

1. Create Account
2. Deposit
3. Withdraw
4. Balance Inquiry
5. View Transactions
6. Exit

Enter your choice:1

Enter Account ID:123

Enter Holder Name:Arun Kumar I A

Enter Initial Balance:Invalid input. Please enter a numeric value.

Enter Initial Balance:1200

Set PIN:1234

Account created successfully!

===== Banking Application =====

1. Create Account
2. Deposit
3. Withdraw
4. Balance Inquiry
5. View Transactions
6. Exit

Enter your choice:2

Enter Account ID:123

Enter PIN:1234

Enter deposit amount:2000

Deposited:2000.0

===== Banking Application =====

1. Create Account
2. Deposit
3. Withdraw
4. Balance Inquiry
5. View Transactions
6. Exit

Enter your choice:4

Enter Account ID:123

Enter PIN:1234

Current Balance:3200.0

===== Banking Application =====

1. Create Account
2. Deposit
3. Withdraw
4. Balance Inquiry
5. View Transactions
6. Exit

Enter your choice:3

Enter Account ID:123

Enter PIN:123

Invalid PIN.

===== Banking Application =====

1. Create Account
2. Deposit
3. Withdraw
4. Balance Inquiry
5. View Transactions
6. Exit

Enter your choice:3

Enter Account ID:123

Enter PIN:1234

Enter withdrawal amount:1000

Withdrawn:1000.0

===== Banking Application =====

1. Create Account
2. Deposit
3. Withdraw
4. Balance Inquiry
5. View Transactions
6. Exit

Enter your choice:4

Enter Account ID:123

Enter PIN:1234

Current Balance:2200.0

===== Banking Application =====

1. Create Account
2. Deposit
3. Withdraw
4. Balance Inquiry
5. View Transactions
6. Exit

Enter your choice:5

Enter Account ID:123

Enter PIN:1234

Deposit of 2000.0 at 2026-09-24T14:10:08.378309

Withdraw of 1000.0 at 2026-09-24T14:11:36.580741

===== Banking Application =====

1. Create Account
2. Deposit
3. Withdraw
4. Balance Inquiry
5. View Transactions
6. Exit

Enter your choice:6

Exiting... Goodbye!

===== Banking Application =====

1. Create Account
2. Deposit
3. Withdraw
4. Balance Inquiry
5. View Transactions
6. Exit

Enter your choice:4

Enter Account ID:123

Enter PIN:1234

Current Balance:3200.0

===== Banking Application =====

1. Create Account
2. Deposit
3. Withdraw
4. Balance Inquiry
5. View Transactions
6. Exit

Enter your choice:3

Enter Account ID:123

Enter PIN:123

Invalid PIN.

===== Banking Application =====

1. Create Account
2. Deposit
3. Withdraw
4. Balance Inquiry
5. View Transactions
6. Exit

Enter your choice:3

Enter Account ID:123

Enter PIN:1234

Enter withdrawal amount:1000

Withdrawn:1000.0

===== Banking Application =====

1. Create Account
2. Deposit
3. Withdraw
4. Balance Inquiry
5. View Transactions
6. Exit

Enter your choice:4

Enter Account ID:123

Enter PIN:1234

Current Balance:2200.0

=== ArrayList Challenge ===

1. Add Item
2. Remove Item
3. Update Item
4. Search Item
5. Display Items
6. Back to Main Menu

Enter your choice:1

Enter item to add:100

=== ArrayList Challenge ===

1. Add Item
2. Remove Item
3. Update Item
4. Search Item
5. Display Items
6. Back to Main Menu

Enter your choice:5

Items:

100

=== ArrayList Challenge ===

1. Add Item
2. Remove Item
3. Update Item
4. Search Item
5. Display Items
6. Back to Main Menu

Enter your choice:2

Enter item to remove:2

=== ArrayList Challenge ===

1. Add Item
2. Remove Item
3. Update Item
4. Search Item
5. Display Items
6. Back to Main Menu

Enter your choice:2

Enter item to remove:100

=== ArrayList Challenge ===

1. Add Item
2. Remove Item
3. Update Item
4. Search Item
5. Display Items
6. Back to Main Menu

Enter your choice:5

Items:

=== ArrayList Challenge ===

1. Add Item
2. Remove Item
3. Update Item
4. Search Item
5. Display Items
6. Back to Main Menu

Enter your choice:1

Enter item to add:100

=== ArrayList Challenge ===

1. Add Item
2. Remove Item
3. Update Item
4. Search Item
5. Display Items
6. Back to Main Menu

Enter your choice:1

Enter item to add:200

=== ArrayList Challenge ===

1. Add Item
2. Remove Item
3. Update Item
4. Search Item
5. Display Items
6. Back to Main Menu

Enter your choice:5

Items:

100

200

=== ArrayList Challenge ===

1. Add Item
2. Remove Item
3. Update Item
4. Search Item
5. Display Items
6. Back to Main Menu

Enter your choice:4

Enter item to search:100

Found!

=== ArrayList Challenge ===

1. Add Item
2. Remove Item
3. Update Item
4. Search Item
5. Display Items
6. Back to Main Menu

Enter your choice:5

Items:

100

200

```
==== ArrayList Challenge ====
1. Add Item
2. Remove Item
3. Update Item
4. Search Item
5. Display Items
6. Back to Main Menu
Enter your choice:5
Items:
100
200
==== ArrayList Challenge ====
1. Add Item
2. Remove Item
3. Update Item
4. Search Item
5. Display Items
6. Back to Main Menu
Enter your choice:4
Enter item to search:100
Found!
==== ArrayList Challenge ====
1. Add Item
2. Remove Item
3. Update Item
4. Search Item
5. Display Items
6. Back to Main Menu
Enter your choice:5
Items:
100
200
==== ArrayList Challenge ====
1. Add Item
2. Remove Item
3. Update Item
4. Search Item
5. Display Items
6. Back to Main Menu
Enter your choice:4
Enter item to search:109
Not found.
```

===== Java Collections Challenge =====

1. ArrayList Challenge
2. HashMap Challenge
3. Queue Challenge
4. Exit

Enter your choice:2

=== HashMap Challenge ===

1. Insert Key-Value
2. Retrieve Value
3. Update Value
4. Check Key
5. Display Entries
6. Back to Main Menu

Enter your choice:1

Enter key:10

Enter value:Arun

=== HashMap Challenge ===

1. Insert Key-Value
2. Retrieve Value
3. Update Value
4. Check Key
5. Display Entries
6. Back to Main Menu

Enter your choice:1

Enter key:20

Enter value:Kumar

=== HashMap Challenge ===

1. Insert Key-Value
2. Retrieve Value
3. Update Value
4. Check Key
5. Display Entries
6. Back to Main Menu

Enter your choice:2

Enter key:10

Arun

=== HashMap Challenge ===

1. Insert Key-Value
2. Retrieve Value
3. Update Value
4. Check Key
5. Display Entries
6. Back to Main Menu

Enter your choice:3

Enter key:10

Enter new value:Sachin

=== HashMap Challenge ===

1. Insert Key-Value
2. Retrieve Value
3. Update Value
4. Check Key
5. Display Entries
6. Back to Main Menu

Enter your choice:5

20 -> Kumar

10 -> Sachin

=== HashMap Challenge ===

1. Insert Key-Value
2. Retrieve Value
3. Update Value
4. Check Key
5. Display Entries
6. Back to Main Menu

Enter your choice:2

Enter key:10

Arun

=== HashMap Challenge ===

1. Insert Key-Value
2. Retrieve Value
3. Update Value
4. Check Key
5. Display Entries
6. Back to Main Menu

Enter your choice:3

Enter key:10

Enter new value:Sachin

=== HashMap Challenge ===

1. Insert Key-Value
2. Retrieve Value
3. Update Value
4. Check Key
5. Display Entries
6. Back to Main Menu

Enter your choice:5

20 -> Kumar

10 -> Sachin

=== HashMap Challenge ===

1. Insert Key-Value
2. Retrieve Value
3. Update Value
4. Check Key
5. Display Entries
6. Back to Main Menu

Enter your choice:4

Enter key:10

Key exists.

=== HashMap Challenge ===

1. Insert Key-Value
2. Retrieve Value
3. Update Value
4. Check Key
5. Display Entries
6. Back to Main Menu

Enter your choice:6

===== Java Collections Challenge =====

1. ArrayList Challenge
2. HashMap Challenge
3. Queue Challenge
4. Exit

Enter your choice:3

=== Queue Challenge ===

1. Add Item
2. Remove Item
3. Peek Front
4. Display Queue
5. Back to Main Menu

Enter your choice:1

Enter item to add:100

=== Queue Challenge ===

1. Add Item
2. Remove Item
3. Peek Front
4. Display Queue
5. Back to Main Menu

Enter your choice:1

Enter item to add:200

=== Queue Challenge ===

1. Add Item
2. Remove Item
3. Peek Front
4. Display Queue
5. Back to Main Menu

Enter your choice:1

Enter item to add:300

=== Queue Challenge ===

1. Add Item
2. Remove Item
3. Peek Front
4. Display Queue
5. Back to Main Menu

Enter your choice:4

Queue:

100

200

300

```
=== Queue Challenge ===
1. Add Item
2. Remove Item
3. Peek Front
4. Display Queue
5. Back to Main Menu
Enter your choice:1
Enter item to add:300
=== Queue Challenge ===
1. Add Item
2. Remove Item
3. Peek Front
4. Display Queue
5. Back to Main Menu
Enter your choice:4
Queue:
100
200
300
=== Queue Challenge ===
1. Add Item
2. Remove Item
3. Peek Front
4. Display Queue
5. Back to Main Menu
Enter your choice:2
Removed: 100
=== Queue Challenge ===
1. Add Item
2. Remove Item
3. Peek Front
4. Display Queue
5. Back to Main Menu
Enter your choice:4
Queue:
200
300
=== Queue Challenge ===
1. Add Item
2. Remove Item
3. Peek Front
4. Display Queue
5. Back to Main Menu
Enter your choice:3
Front: 200
=== Queue Challenge ===
1. Add Item
2. Remove Item
3. Peek Front
4. Display Queue
5. Back to Main Menu
Enter your choice:5
===== Java Collections Challenge =====
1. ArrayList Challenge
2. HashMap Challenge
3. Queue Challenge
4. Exit
Enter your choice:
```

=== Queue Challenge ===

1. Add Item
2. Remove Item
3. Peek Front
4. Display Queue
5. Back to Main Menu

Enter your choice:1

Enter item to add:200

=== Queue Challenge ===

1. Add Item
2. Remove Item
3. Peek Front
4. Display Queue
5. Back to Main Menu

Enter your choice:1

Enter item to add:300

=== Queue Challenge ===

1. Add Item
2. Remove Item
3. Peek Front
4. Display Queue
5. Back to Main Menu

Enter your choice:4

Queue:

100

200

300

=== Queue Challenge ===

1. Add Item
2. Remove Item
3. Peek Front
4. Display Queue
5. Back to Main Menu

Enter your choice:2

Removed: 100

=== Queue Challenge ===

1. Add Item
2. Remove Item
3. Peek Front
4. Display Queue
5. Back to Main Menu

Enter your choice:4

Queue:

200

300

===== Library Management System =====

1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Remove Book
7. View Issue Records
8. Exit

Enter your choice:1

Enter Book ID:123

Enter Title:Wings of Fire

Enter Author:Dr. APJ Abdul Kalam

Book added successfully!

===== Library Management System =====

1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Remove Book
7. View Issue Records
8. Exit

Enter your choice:2

123 - Wings of Fire by Dr. APJ Abdul Kalam [Available]

===== Library Management System =====

1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Remove Book
7. View Issue Records
8. Exit

Enter your choice:3

Search by (1) ID or (2) Title:1

Enter ID:123

123 - Wings of Fire by Dr. APJ Abdul Kalam [Available]

===== Library Management System =====

1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Remove Book
7. View Issue Records
8. Exit

Enter your choice:4

Enter Book ID to issue:123

Enter User ID:123

Enter User Name:Arun

Book issued to Arun

===== Library Management System =====

1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Remove Book
7. View Issue Records
8. Exit

Enter your choice:5

Enter Book ID to return:111

No active issue record found for this book.

===== Library Management System =====

1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Remove Book
7. View Issue Records
8. Exit

Enter your choice:5

Enter Book ID to return:123

Book returned successfully!

===== Library Management System =====

1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Remove Book
7. View Issue Records
8. Exit

Enter your choice:7

Book: Wings of Fire, User: Arun, Issued: 2026-09-24, Returned: 2026-09-24

===== Library Management System =====

1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Remove Book
7. View Issue Records
8. Exit

Enter your choice:5

Enter Book ID to return:123

Book returned successfully!

===== Library Management System =====

1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Remove Book
7. View Issue Records
8. Exit

Enter your choice:7

Book: Wings of Fire, User: Arun, Issued: 2026-09-24, Returned: 2026-09-24

===== Library Management System =====

1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Remove Book
7. View Issue Records
8. Exit

Enter your choice:6

Enter Book ID to remove:123

Book removed successfully!

===== Library Management System =====

1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Remove Book
7. View Issue Records
8. Exit

Enter your choice:8

Exiting... Goodbye!

===== Library Management System =====

1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Remove Book
7. View Issue Records
8. Exit

Enter your choice:1

Enter Book ID:123

Enter Title:Wings of Fire

Enter Author:Dr. APJ Abdul Kalam

Book added successfully!

===== Library Management System =====

1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Remove Book
7. View Issue Records
8. Exit

Enter your choice:3

Search by (1) ID or (2) Title:2

Enter Title:Wings of Tyre

Book not found.

===== Library Management System =====

1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Remove Book
7. View Issue Records
8. Exit

Enter your choice:3

Search by (1) ID or (2) Title:2

Enter Title:Wings of Fire

123 - Wings of Fire by Dr. APJ Abdul Kalam [Available]